

JURONG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATIONS
Higher 2

CANDIDATE
NAME

CLASS

BIOLOGY

9648/03

Applications Paper and Planning Question

13 September 2016

Paper 3

2 hours

Additional Materials: Answer Booklet/Paper

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1	
2	
3	
4	
5	
Total	

This document consists of **13** printed pages and **1** blank page.

[Turn over]

BLANK PAGE

Answer **all** the questions in this section.

- 1 Fig. 1.1 shows the plasmid pOLUC. Plasmid pOLUC is a cloning vector used in creating recombinant DNA. It contains two selectable marker genes, luciferase gene and ampicillin resistance gene, Amp^R. A multiple cloning site (MCS) is located within the luciferase gene. Luciferase gene, when expressed, hydrolyses luciferin which results in a glowing bacterial colony.

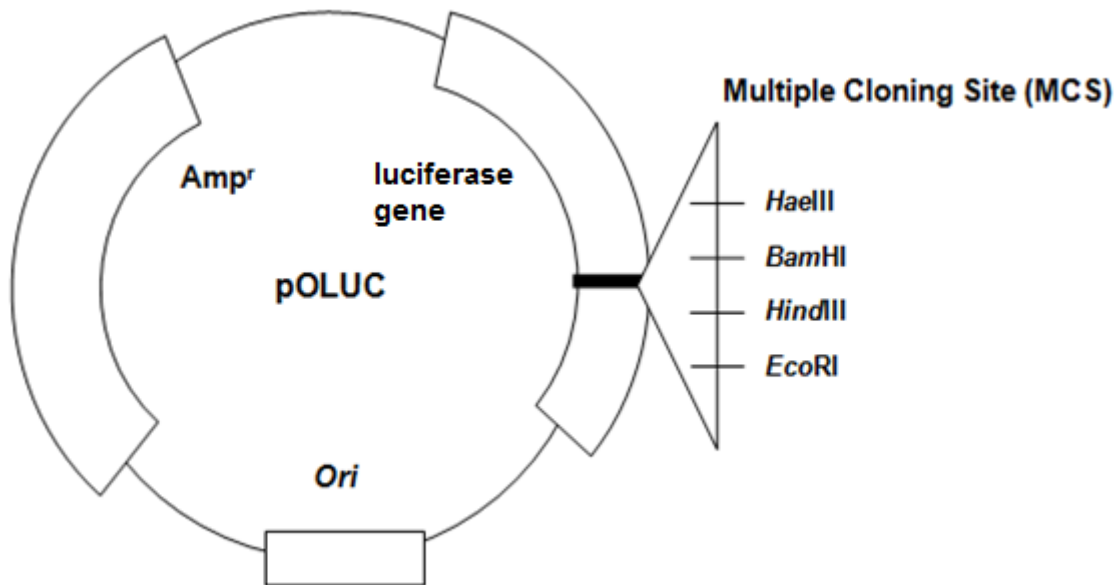


Fig. 1.1

(a) Describe and explain one property of pOLUC that allows it to be used as DNA cloning vector. [2]

1. All plasmids contain an origin of replication;;
2. so that the vector and the inserted gene of interest can replicate independently to produce multiple copies within the host cell;;
3. Plasmids usually contain (at least 2) selectable/ genetic markers;;
4. e.g. antibiotic resistant genes which confer certain well-defined phenotypes on the host organism and enable selection of bacteria;;
5. Plasmids also possess restriction sites;;
6. which can be cut by restriction enzymes for insertion of gene of interest (to be cloned);;

Fig. 1.2 shows the target DNA sequence that will be inserted into the MCS of pOLUC.

5' – GGATCCATCCCGATCGATCCTTTAAGCTAGCTAGGATCC – 3'
 3' – CCTAGGTAGGGCTAGCTAGGAAATTCGATCGATCCTAGG – 5'

Fig. 1.2

Fig. 1.3 shows the types of restriction enzymes that can be used to cleave pOLUC at the Multiple Cloning Site (MCS) shown in Fig. 1.1.

Restriction enzyme	Specific target base sequence of DNA
<i>HaeIII</i>	5' ... G G C ... 3' 3' ... C C G ... 5'
<i>BamHI</i>	5' ... G G A T C C ... 3' 3' ... C C T A G G ... 5'
<i>HindIII</i>	5' ... A A G C T T ... 3' 3' ... T T C G A A ... 5'
<i>EcoRI</i>	5' ... G A A T T C ... 3' 3' ... C T T A A G ... 5'

Fig. 1.3

(b) Describe the natural function of a restriction enzyme. [2]

1. Protect the bacteria from attack by bacteriophages / viruses by recognising, binding & cutting viral genome/foreign DNA that entered the bacterial cell;;
2. via hydrolyzing phosphodiester bonds between adjacent nucleotides;;

(c) State the restriction enzyme used to cleave the target DNA sequence before it is inserted into vector pOLUC. Give a reason for your choice. [2]

1. *BamHI*;;
2. Presence of restriction site GGATCC (and sticky ends are generated after digestion with *BamHI*);;

(d) Outline the steps in producing a recombinant plasmid. [3]

1. The target sequence and plasmid are cleaved/ digested by *BamHI* / same restriction enzyme to give sticky ends/ staggered ends;;
2. The target sequence and the plasmid are then mixed together to allow sticky ends/ staggered ends to anneal by hydrogen bonding / complementary base pairing;;
3. DNA ligase is added to form phosphodiester bonds /seal the nicks between the target sequence and the plasmid;;

(e) Explain how bacteria with recombinant plasmids are selected from bacteria that are transformed. [5]

1. The bacteria are first plated onto nutrient agar plates containing the ampicillin and luciferin;;
2. Bacterial colonies containing recombinant plasmid grow but do not glow on the ampicillin and luciferin plate;;
3. Transformed bacteria (with recombinant and non-recombinant plasmid) will grow as they are resistant to ampicillin / contain the ampicillin resistance gene;;
4. Gene of interest is inserted into the luciferin gene leading to insertional inactivation therefore cells do not glow;;
5. Cells containing non-recombinant plasmid will glow on luciferin plate as the luciferase gene is intact, hence, able to produce luciferase to generate light from luciferin/ hydrolyse luciferin to produce light;;

(f) Suggest why replica plating was not necessary in this experiment. [1]

1. Bacteria with recombinant plasmids can be differentiated by their ability to glow and they do not die in the presence of ampicillin;;

[Total: 15]

- 2 Cystic fibrosis (CF) is a genetic disorder caused by mutations in the gene that code for the CF transmembrane conductance regulator (CFTR) that contributes to regulation of ion flux at the membrane of epithelial cells.

A DNA sequence, KM-19 found within 1 centiMorgan of the CFTR gene has been identified. This segment exhibits restriction fragment length polymorphism (RFLP) that can be used as a RFLP marker for the genetic screening of CF.

DNA samples containing the CFTR gene and KM-19 from four individuals were treated with restriction enzyme, *EcoRI* and subjected to gel electrophoresis.

(a) Outline why gel electrophoresis separates DNA fragments. [3]

1. When placed in an electric field, negatively-charged DNA molecules will move towards the anode (positively charged electrode);;
2. As electrophoresis is performed in an agarose gel which consists of a complex network of pores, DNA fragments of different sizes in a given sample can be separated based on their different rates of migration through the gel.
3. Longer DNA fragments will have more difficulty (encounter more resistance) in moving through the pores in the gel and will therefore move slower at a lower rate and be found closest to the well (or vice versa);;

Fig. 2.1 shows the positions of various restriction fragment of KM-19 and a probe that was used to detect the polymorphism. RFLP variant 1 of KM-19 was associated with the normal CFTR allele while RFLP variant 2 of KM-19 was linked to the mutant CFTR allele.

It was observed that individual C does not show the expected band pattern for affected individual with CF.

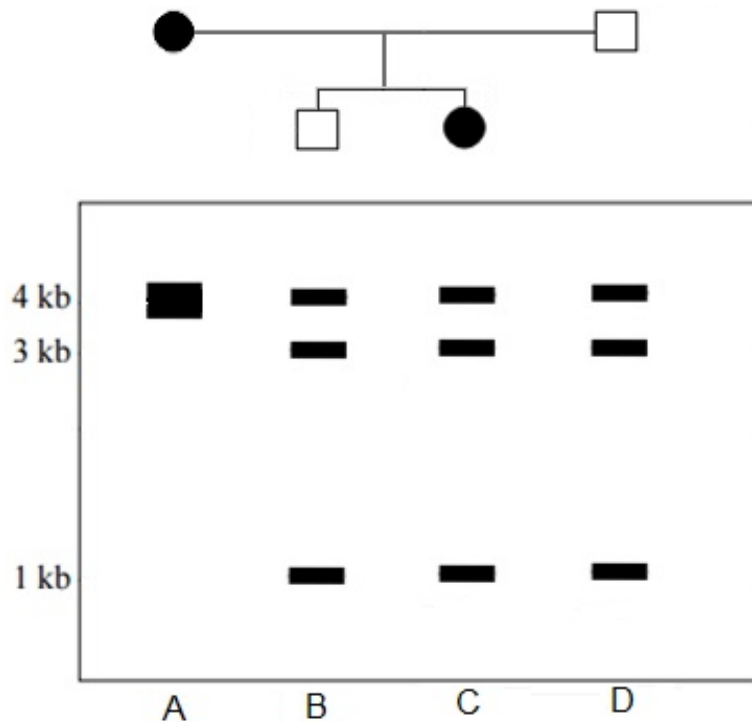
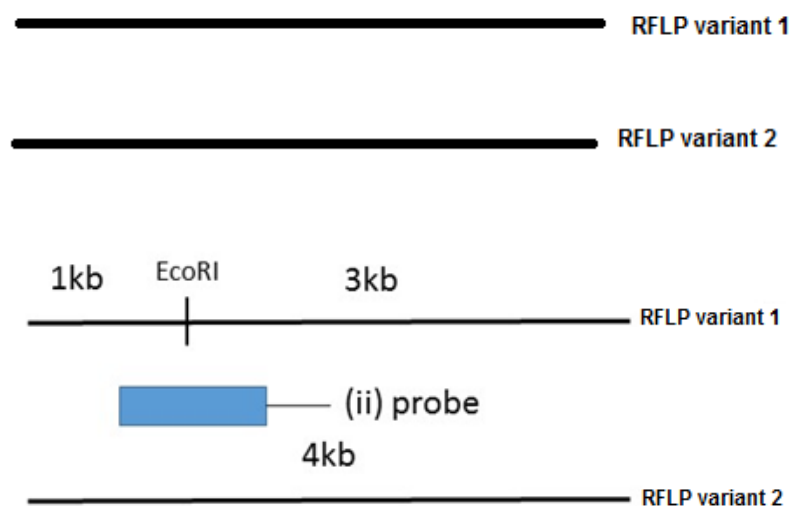


Fig. 2.1

(b) With reference to Fig. 2.1,

- (i) indicate the restriction site(s) of *EcoRI* and DNA fragment sizes on RFLP variants 1 and 2 of KM-19; [1]
- (ii) draw on the restriction map where the probe binds. [1]



(c) Explain why genetic markers like KM-19 can be used to detect the presence of disease-causing alleles. [2]

1. These genetic markers are polymorphic/different nucleotide sequences that can be recognized and cut differently by the same restriction enzyme;;
2. They are tightly linked to the disease-causing alleles on the same chromosome and are likely to be inherited together as one unit with the disease-causing allele;;

(d) Suggest an explanation for the discrepancy observed in C regarding the band pattern and occurrence of CF. [3]

1. C has both RFLP variant 1 and 2 which suggests that she should be a heterozygote/has 1 copy of normal CFTR allele and 1 copy of mutant CFTR allele;;
2. However, C suffers from CF which meant that she should have 2 copies of mutant CFTR recessive alleles which are linked to RFLP variant 2;;
3. There could be a mutation in RFLP variant 2 resulting in creation of a restriction site OR
4. (During the formation of gamete in the father), crossing over occurred at a region between KM-19 and CF locus causing RFLP variant 1 to become associated with mutant allele;;

(e) In the absence of gel electrophoresis and nucleic acid hybridisation, outline a possible method in detecting RFLP variant 2 of KM-19. [1]

1. Design primers that are complementary to the RFLP variant 2 of KM-19 and use PCR to synthesise variant 2 of KM-19 and mutant CFTR;;

[Total: 11]

- 3 Severe combined immunodeficiency (SCID) is a genetic disorder whereby the affected individuals have non-functional immune system with increased vulnerability to infections. The most common form of SCID is X-linked SCID, caused by mutation of interleukin-2 receptor gamma (IL2RG) gene on X chromosome.

Fig. 3.1 shows interleukin-2 receptor that is involved in JAK-STAT signalling pathway in the T cells of a healthy individual.

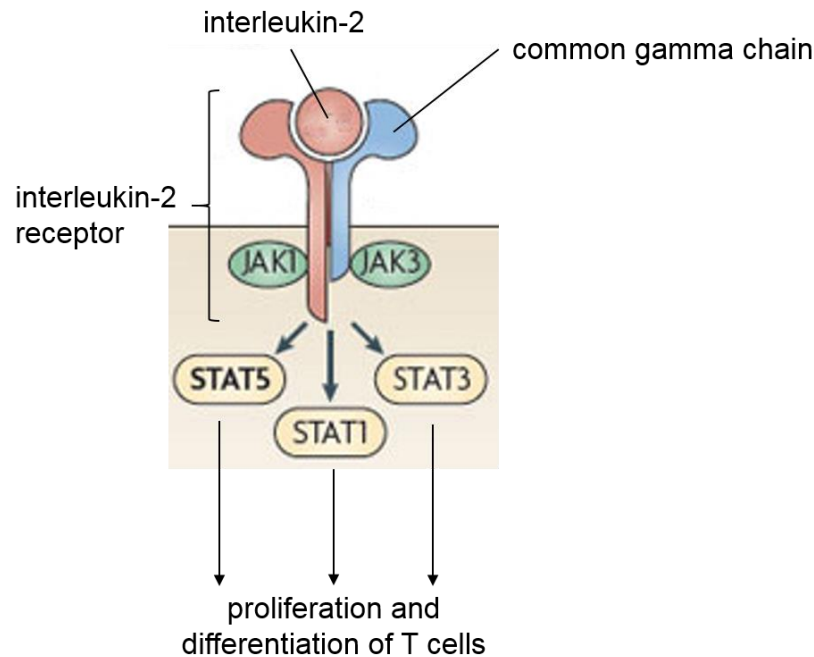


Fig. 3.1

(a) With reference to Fig. 3.1 and the information above, describe the mutation that results in X-linked SCID. [3]

1. Mutation of IL2RG gene results in the conformational change of common gamma chain on interleukin receptor;;
2. Which prevents interleukin-2 from recognising and binding to the receptor;;
3. No signal transduction via JAK-STAT signalling pathway and T cells cannot undergo proliferation and differentiation;;
4. Low number/absence of T cells prevent communication with/activation of B cells and thus, non-functional immune system;;

In 2002, two young individuals with X-linked SCID were treated with retroviral-based gene therapy on their isolated blood stem cells.

(b) Outline how a retroviral vector delivers normal functional IL2RG allele into the blood stem cells. [3]

1. RNA copies of normal functional IL2RG allele are inserted into inactivated, non-pathogenic retrovirus;;
2. The blood stem cells are infected with retroviruses which introduce the normal functional allele/releases genetic material into the cells;;
3. RNA are reverse transcribed into the DNA of normal functional IL2RG allele which is subsequently integrated into genome of blood stem cells by integrase;;

Fig. 3.2 shows the average lymphocyte count (A) and antibody level (B) of two individuals that underwent retroviral-based gene therapy treatments. A control (individual without X-linked SCID) is used as the reference of average lymphocyte count and antibody level in a healthy person.

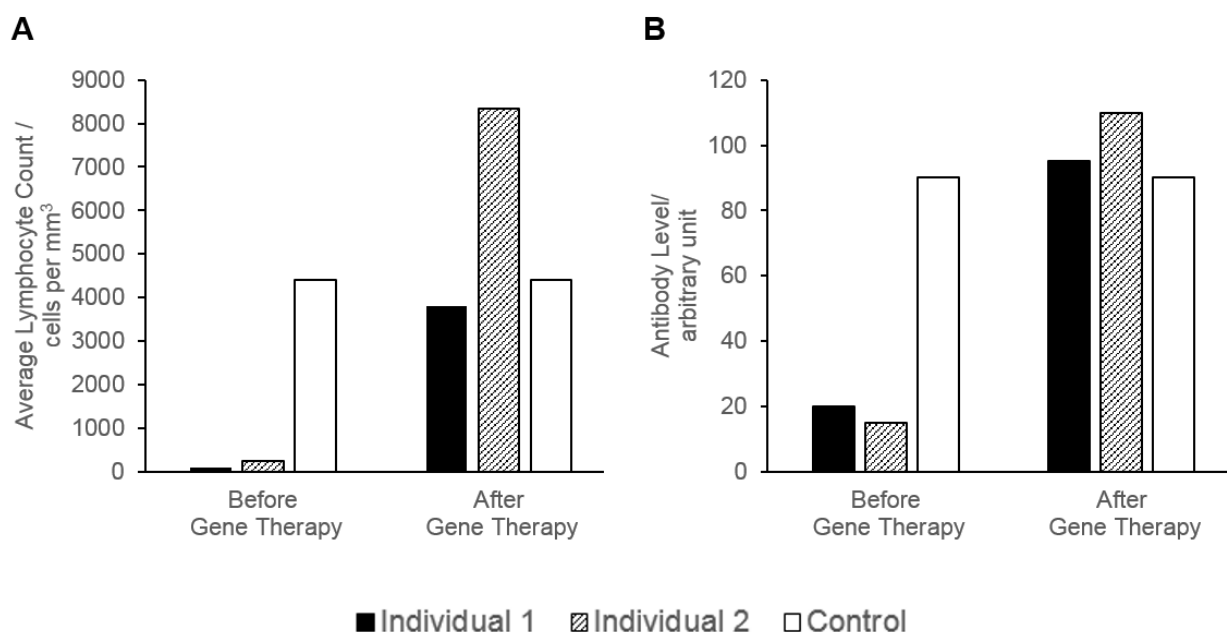


Fig. 3.2

(c) With reference to Fig. 3.2A and Fig. 3.2B,

(i) account for the effectiveness of gene therapy on Individual 1; [4]

1. Gene therapy increases the average lymphocyte count in Individual 1 from 100 to 3800 cells per mm³;;
2. Due to the blood stem cells proliferating and differentiating into new T cells and B cells with normal functional common gamma chain/interleukin-2 receptor;;
3. Gene therapy also increases the antibody level in Individual 1 from 20 to 95 arbitrary unit;;
4. Communication between B cells and T cells allows for (the activation of B cells and thus) the production of antibodies;;

(ii) suggest and explain the potential safety concern in Individual 2. [2]

1. Individual 2 may develop cancer due to excessive proliferation of lymphocytes / average lymphocytes count in Individual 2 is nearly twice as high as the control;;
2. Random insertion of the allele within the DNA/genome by retrovirus resulting in insertional mutagenesis of tumour suppressor genes/ proto-oncogene;;

(d) Discuss one ethical and one social consideration for the use of gene therapy. [2]

Social Concern (any one)

1. Society may be less accepting of people who have mild disorders / genetic diseases or are less-able leading to possible discrimination and practice of eugenics;;
2. Attempts at gene therapy are exorbitantly expensive leading to social class discrimination with the rich having better accessibility to treatments as compared to the poor / genetic enhancements creating an advantage for those who can afford the treatment;;

Ethical Concern (any one)

3. Safety of gene therapy to humans: Long term consequences have not been fully established due to inadequate understanding of biological interactions of vector and host / In the event of undesirable outcomes brought about by the treatment, it is unclear who should be held responsible / There are questions over who should participate in a human trial – the terminally ill with no treatment options or anyone who understands the potential risk but is a willing patient;;
4. There concerns about the protection of privacy and confidentiality of medical information of patients involved in clinical trials or those who have gone for such treatments as this could have implications on their medical insurance coverage / future employability;;
5. Difficulty in determining which condition is normal and which are considered as a disability/disorder: It may become increasingly difficult to define which gene traits are considered to be disability/disorder (e.g. dwarfism disorder/ achondroplasia) versus those that should be classified as physical, mental, or psychological traits (e.g. short stature);;
6. Gene therapy for non-health related enhancement: There are concerns that gene therapy may be abused to alter human traits not associated with disease / for genetic enhancement purposes e.g. having taller height;;
7. Altering one's genes and changing the way one was created goes against many religious and cultural beliefs;;

[Total: 14]

4 Planning Question

The yield of crop plants such as the onion, *Allium cepa*, is reduced by the presence of relatively high concentrations of salts, particularly sodium chloride, in the soil. There are, however, salt-tolerant varieties of onion, which can be grown successfully in such places. One possible explanation for this is that the fluid in the cell vacuoles of plants of the tolerant varieties differs in solute potential from that of non-tolerant varieties.

To measure the solute potential of the cell vacuole, one can make use of the concept of incipient plasmolysis. Incipient plasmolysis is the point where 50% of plant cells are plasmolysed.

Plan an investigation to find out whether the solute potentials of the cell vacuoles of epidermal cells of the bulbs of salt-tolerant onions differ from those of their non-tolerant counter parts.

You are provided with:

- 1.0 mol dm⁻³ sodium chloride solution
- two varieties of onion, one salt-tolerant and the other non-salt tolerant
- distilled water
- beakers of various sizes
- glass specimen tubes
- microscopic slides and coverslips
- measuring cylinders of various sizes
- dropping pipettes
- light microscope with low (x100), medium (x200) and high (x400) magnification and built in illumination
- stopwatch
- scalpel
- forceps
- glass rods for stirring
- mounted needles
- iodine in potassium iodide solution as a stain

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it,
- be illustrated by relevant diagram(s), if necessary,
- identify the independent and dependent variable,
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and reliable as possible,
- include layout of results tables and graphs with clear headings and labels,
- use the correct technical and scientific terms,
- include reference to safety measures to minimise any risks associated with the proposed experiment.

[Total: 12]

Mark Scheme

1. Theoretical consideration or rationale of the plan to justify the practical procedure.
 - a) tolerant variety is caused by higher concentration of external solution
 - b) osmosis
2. Identification of variables – independent and dependent variables.
3. Description of how dependent variable could be measured
4. Identification of other variables to be kept constant (name 2)
5. Dilution table to make up 5 different concentrations of NaCl
6. Description of how onion sample can be stained before viewing under microscope
7. Description of how to determine plasmolysis
8. Relevant clearly labelled diagram
 - Preparing slide for microscopy OR
 - How a plasmolysed cell looks like
9. Replicates and at least two repeats.
 - Step 11 & 14 OR
 - Table showing replicates and step 14
10. Table with appropriate column heading and showing average
 - Column for concentration of NaCl, average % plasmolysis for 2 varieties of onion
11. Graph with appropriate axis and showing that non-salt tolerant variety reaches 50% plasmolysis at a lower concentration than salt-tolerant
12. Safety precautions / risk hazards identified
13. The correct use of technical and scientific terms

[12 marks]

Variables [2]

The independent variable is concentration / solute potential of external solution/NaCl while dependent variable is the percentage of cells plasmolysed.

Other variables to be kept constant (name 2) [4]

- Immersion time
- Volume of immersion solution
- Size of onion sample

Prediction of the likely outcome of the experiment [1a]

Tolerant variety has a lower (more negative) solute potential than non-tolerant variety. Incipient plasmolysis in tolerant variety is caused by higher concentration of external solution than in case of cells of the non-tolerant variety.

Explain the likely outcome of the experiment [1b]

Water moves by osmosis from higher water potential to lower water potential/ towards the more negative solute potential therefore when external solute potential is low (as in a concentrated salt solution) the internal solute potential must be even lower (as in more concentrated internal solution)

Explanation of how dependent variable can be measured [3]

Determine the external concentration / solute potential which results in incipient plasmolysis / 50% plasmolysis in cells of each variety.

The solute potential of the cells of each variety of onion is equivalent to that of the solution which causes 50% plasmolysis because at this point both external and internal solute potential are equal.

Procedure (includes control and reliability of results)

1. Label 5 specimen tubes A-E.
2. Prepare 10 cm³ of varying NaCl concentration as indicated in the table. [5]

Tubes	Concentration of NaCl solution / mol dm ⁻³	Volume of 1.0 mol dm ⁻³ NaCl solution / cm ³	Volume of distilled water / cm ³
A	0.2	2.0	8.0
B	0.4	4.0	6.0
C	0.6	6.0	4.0
D	0.8	8.0	2.0
E	1.0	10.0	0.0

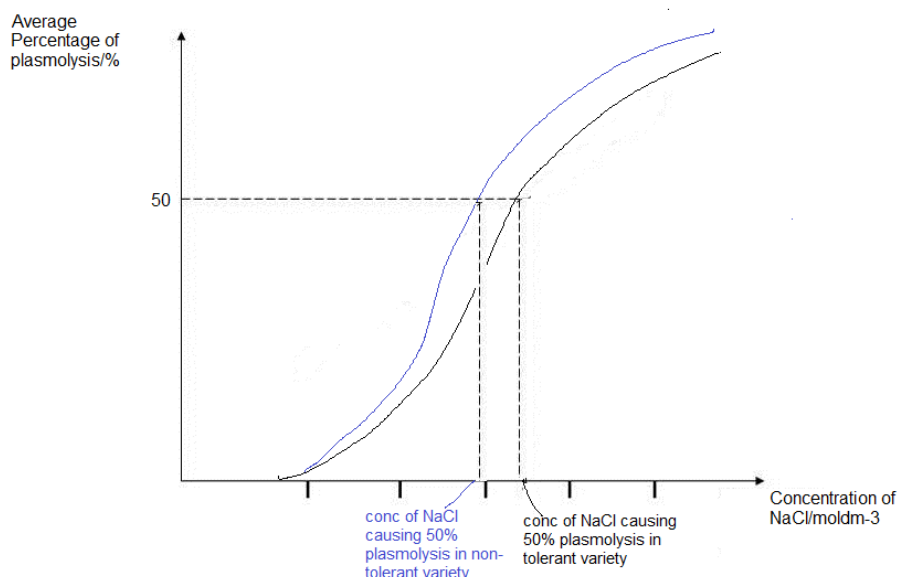
3. Remove epidermis from inner surface of onion scale with scalpel.
4. **Use the scalpel to cut 6 pieces of 2 cm by 1 cm epidermal tissue from salt-tolerant onion. [4]**
5. **Using the forceps, put the tissue into 5 cm³ of A in a glass specimen tube. [4]**
6. **Incubate the onion samples for 15 minutes as timed using a stopwatch. [4]**
7. Remove the onion epidermal tissue and place it on a microscopic slide.
8. Add a drop of iodine in potassium iodide solution using a dropper. [6]
9. Cover the sample with a coverslip using mounting needle and place the slide onto the microscope.
10. **Count the number of cells undergoing plasmolysis in the first 30 cells seen in this field of view. [7]**

11. Repeat step 10 by viewing other part of the onion sample for 2 more readings. [9]
12. Calculate percentage of plasmolysed cells.
13. Repeat steps 5-12 for the other concentrations of NaCl solution (B-E).
14. Repeat steps 1-13 twice. [9]
15. Plot a graph of average % plasmolysis against concentration of NaCl solution for both varieties of onion.

Table [9,10]

Concentration of NaCl/ mol dm ⁻³	No. of salt-tolerant cells plasmolyzed				Average % plasmolysed cells in salt tolerant sample	No. of non-tolerant of cells plasmolyzed				Average % plasmolysed cells in non-tolerant sample
	Reading 1	Reading 2	Reading 3	Average		Reading 1	Reading 2	Reading 3	Average	
0.0										
0.2										
0.4										
0.6										
0.8										
1.0										

Graph [11]



Risk and precaution [12]

Put scalpel/mounting needle away after use to avoid cutting oneself.
Prevent iodine from coming into contact with eyes by wearing safety goggles.

Free-response question

Write your answers to this question on the separate answer paper provided.

Your answers:

- should be illustrated by large, clearly labelled diagrams, where appropriate.
- must be in continuous prose, where appropriate.
- must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 5 **(a)** Describe features of embryonic stem cells and adult stem cells that distinguish them from each other. [6]

Features	Embryonic stem cells	Adult stem cells
Potency	1. Pluripotent/exhibit pluripotency;;	2. Multipotent/exhibit multipotency;;
	3. Have ability to differentiate into almost any cell type to form any organ or type of cell and so are so are pluripotent and multipotent;;	4. Have the ability to differentiate into a limited range of cell type and so are not pluripotent or totipotent but are multipotent;;
Location	5. Undifferentiated and unspecialized cell found at inner cell mass of blastocyst;;	6. Undifferentiated and unspecialized cell found among differentiated cells in a tissue or organ;;
Function	7. Have the ability to differentiate into almost any cell types in the body <u>except extra-embryonic membrane/trophoblast</u> to form any organ or type of cell;; 8. The inner cell mass can give rise to the derivatives of the three germ layers: endoderm, mesoderm, and ectoderm which eventually form all the highly specialized cells needed to produce an adult organism. This means that embryonic stem cells have the potential to make almost all cell types in the body;;	9. To maintain and repair the specific tissue where they reside by replacing worn out or damaged cells;;

(b) Outline the procedure for propagating plants by tissue culture technique. [6]

1. Explant containing meristematic cells is selected from shoot tip/root tip/small pieces of leaf and stem;;
2. and is sterilized using dilute sodium hypochlorite solution;;
3. The explant is then aseptically transferred to culture vessels containing aseptic culture medium that contains nutrients and plant growth regulators;;
4. Auxin and cytokinin in equal ratio is added to the nutrient agar to stimulate the cells of the explant to divide by mitosis to form a callus;;
5. As callus increases in size, it can be subcultured into many calli by placing them in separate culture vessels containing culture medium with plant growth regulators;;
6. The cells of a callus can be induced to either proliferate or differentiate into particular tissues e.g. shoot or root by varying the ratio of the various plant growth regulators / Low level of cytokinin and high level of auxin triggers formation of roots growth / High level of cytokinin and low level of auxin triggers shoot growth / Auxin and gibberellin stimulates cell differentiation;; (any one e.g.)
7. With differentiation of cells of callus into particular tissues, a plantlet can form;;
8. Plantlets are then allowed to acclimatized before transfer to soil to grow into a whole plant;;

- (c) Explain, with a named example, how genetic engineering improves the yield of crops. Include, in your explanation, two concerns associated with this genetically modified crop. [8]

Definition of genetic engineering [1]

1. Genetic engineering or genetic modification (GM) refers to a special set of technologies that alter the genetic material of animals, plants or bacteria by introducing genes from different species into target organisms;;

Bt Corn [5]

2. Named crop: Bt corn;;
3. Bt toxin genes from *Bacillus thuringiensis* is inserted into corn genome;;
4. Bt toxin/insecticidal crystalized proteins (ICPs) are lethal to insect larvae when they are ingested;;
5. Problem: Larvae of *Lepidoptera* species that ingest corn causing reduced yield for corn crops for farmers;;
6. Solution: The protein toxin produced kills the larvae of *Lepidoptera* species, including the European corn borer. Larvae which ingest the toxin die because of damage caused to the gut. Hence these larvae are unable to cause further damage to crops, farmers will have higher crop yield;;
7. This allows farmers to concentrate resources on other aspects like fertilizers, improved farming practices that result in further improved yield;; (any one example)

Two concerns [2]

8. Bt toxin may cause allergies in some human when Bt corn is consumed;;
9. Bt toxin may be toxic to close relatives of targeted pests/beneficial insects, leading to reduced numbers or extinction of that organism;;
10. Pollen drift - Pollen of Bt corn may be carried on the wind or by animals and pollinates nearby non-GM crops or wild plants. Mixing GM crops with non-GM products confounds labelling attempts;;
11. Tampering with nature by mixing genes among species – Bt toxin genes are sourced from *Bacillus thuringiensis*;;

[Total: 20]